

An Open Set Recognition for SAR Targets Based on Encoding-Conditional Decoding Network with Reject Threshold Adaptation

Yiming Li, Lan Du, *Senior Member, IEEE*, Jian Chen, Jialun Song, Zilin Wang, Yuchen Guo

Abstract—Current synthetic aperture radar (SAR) automatic target recognition (ATR) methods primarily focus on closed set recognition, i.e., they typically assume that the training set covers all target classes that appear in the test set. However, this assumption may not hold in practical scenarios, where it's inevitable to encounter high-value targets whose classes are absent from the training set. In this article, we propose an open set recognition (OSR) based on the encoding-conditional decoding network with reject threshold adaptation (ECTA) for SAR targets. The ECTA consists of four parts: 1) the encoding; 2) classification; 3) conditional decoding; and 4) reject threshold calculation modules. The encoding module learns the latent representation of the training samples, which is then fed to the classification part to predict the class labels. The conditional decoding module is trained to produce matching reconstructed samples by combining the latent representation with the label corresponding to the target class while producing nonmatching reconstructed samples by combining the latent representation with the label corresponding to a random other target class. Through the proposed reconstruction strategy, the conditional decoding module gives an approximate description of the known and unknown class reconstruction error distributions, building the foundation for the reject threshold calculation module. The reject threshold calculation module fits the extreme value distributions of matching and nonmatching reconstruction errors based on extreme value theory (EVT) and finally obtains satisfactory reject thresholds adaptively by minimizing the probability of false positives and false negatives. Experiments conducted on two measured datasets indicate that ECTA outperforms state-of-the-art OSR methods, such as DIAS (+0.01%~7.36% and +2.1% in F1-score on MSTAR against varying openness and FUSAR-Ship respectively).

Index Terms—Synthetic aperture radar (SAR), automatic target recognition (ATR), conditional decoding, extreme value theory (EVT), reject threshold adaptation.

I. INTRODUCTION

SYNTHETIC aperture radar (SAR) has rapidly advanced due to its exceptional imaging capabilities. It remains unrestricted by constraints related to time, lighting, and weather conditions [1, 2]. As SAR sensor technology progresses and the volume of SAR images increases, the field of SAR image

interpretation [3–12] has captured substantial interest and emerged as a prominent and vibrant field of research. Within this field, a crucial subfield is SAR automatic target recognition (ATR), the significance of which lies in its indispensable role in identifying the SAR targets. This process is paramount for precise image interpretation and in-depth analysis, contributing significantly to the field of remote sensing.

With the improvement of computing power, deep learning-based SAR ATR methods have demonstrated considerable superiority over traditional machine learning approaches, emerging as the predominant methodologies. Most SAR ATR methods [8–15] with deep learning concentrate on how to extract more distinguished target features to improve recognition performance. For example, some researchers [8–10] modify the architecture of convolutional neural networks (CNNs) to make them more suitable for SAR ATR tasks. Besides, some recent studies [11–15] successfully integrate the physical scattering model of SAR into the CNNs, such as attributed scattering center (ASC) model or electromagnetic scattering model. These methods further improve recognition performance and achieve a higher level of physical interpretability.

However, despite existing SAR ATR methods have achieved excellent recognition performance, the majority of them are designed under the closed set assumption: they assume target classes in the test set have all appeared and been learned in the training stage. In fact, this assumption does not always hold in real-world scenarios. For example, due to confidentiality considerations, many high-value targets are difficult to observe in advance in combat scenarios. Therefore, it's impossible to establish a complete training set covering all target classes. In practice, the presence of unknown classes not included in the training set is a common occurrence. Taking the target classes shown in Fig. 1(a) as examples, the closed set SAR ATR methods learn the half-open decision boundaries according to the training samples, and recognize unknown targets as the known classes as shown in Fig. 1(b), which greatly limits the applications of current SAR ATR technologies [16]. Recently, open set recognition (OSR) has been introduced as a solution for addressing recognition challenges arising from the presence of unknown classes. In detail, the OSR methods aim to recognize known classes and reject unknown classes simultaneously by learning the closed decision boundary for each known class as shown in Fig. 1(c). Targets inside the closed decision boundaries will be recognized as corresponding known classes, while targets outside the closed decision boundaries will be rejected as unknown classes. Therefore,

This work was supported in part by the National Science Foundation of China under Grant U21B2039, 62201431 and 62201433, the Fundamental Research Funds for the Central Universities QTZX23067 and ZYTS24130. (Corresponding author: Lan Du.)

Yiming Li, Lan Du, Jian Chen, Jialun Song and Zilin Wang are with the National Key Laboratory of Radar Signal Processing, Xidian University, Xi'an 710071, China. (e-mail: liyiming0823@163.com, dulan@mail.xidian.edu.cn, jianc@xidian.edu.cn, xd_sjl@163.com, zlwang_1997@stu.xidian.edu.cn).

Yuchen Guo is with the Academy of Advanced Interdisciplinary Research, Xidian University, Xi'an 710071, China. (e-mail: ychguo@xidian.edu.cn)

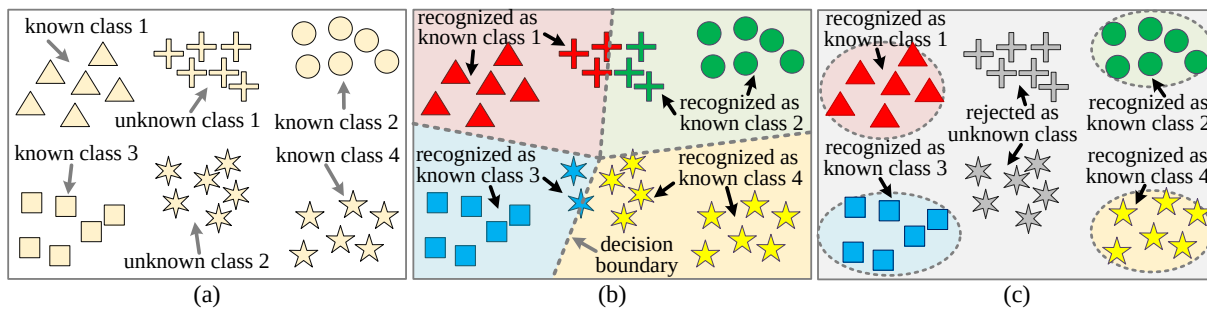


Fig. 1. Comparison between closed set recognition methods and open set recognition methods. Fig. 1(a) shows the distribution sketch map of data from known and unknown class targets in the open set scenario. Fig. 1(b) shows the half-open decision boundaries learned by the closed set recognition methods that recognize unknown class targets as known classes. Fig. 1(c) shows the closed decision boundaries learned by the open set recognition methods that recognize known class targets and reject unknown class targets simultaneously.

delving into OSR technologies holds significant importance for real-world applications of SAR ATR [17].

OSR tasks involve rejecting test samples of unknown classes that do not appear in the training stage while recognizing test samples of known classes, which is somewhat similar to out-of-distribution (OOD) detection [18]. However, there are differences in the definition of OOD data between OOD detection and the OSR tasks [19–21]: In the OSR tasks, OOD data refers to samples with semantic differences from the training samples (inconsistent label space), while in OOD detection, OOD data refers to samples with statistical differences from the training samples (inconsistent data distribution). This paper focuses on rejecting the high-value targets whose classes are absent from the training set, such as new types of vehicle targets. Therefore, this paper should be classified under the field of OSR.

Numerous OSR methods have merged in the realm of computer vision [22]. Recently, scholars have begun applying OSR techniques to SAR ATR [17, 23–26]. Generally, these OSR methods can be roughly divided into two groups: those based on generative model and those based on discriminative model.

The key to OSR methods based on generative model is to generate unknown class examples, and use both known class training examples and generated unknown class examples to jointly learn a network with $K + 1$ classification output nodes (K represents the number of target classes included in the training set and the $K + 1$ node corresponds to the unknown class). A straightforward idea is to generate fake unknown class examples via a generative adversarial network (GAN) [27–31] or mixup strategy [13, 23]. Ge et al. [27] first apply the generative model to the OSR tasks. In particular, they modify the training procedure of conditional GAN and mix several prior known class distributions to generate plausible unknown class samples. Neal et al. [28] propose OSRCI, a method employing an encoder-decoder GAN. This model is devised to generate samples representing unknown classes, exhibiting similarities to known class samples while remaining distinct from any known classes. Similarly, Jo et al. [29] employ GAN to generate samples representing unknown classes. Moon et al. [30] propose DIAS, a technique that utilizes GAN to produce unknown class examples, spanning

various levels of complexity. Li et al. [31] mitigates the mode collapse problem of the generator by constraining the diversity among generated samples, thereby producing relatively diverse generated samples to represent unknown classes. Zhou et al. [13] propose PROSER that mixes up deep features of different target classes from the training set to mimic unknown class patterns. Similarly, Geng et al. [23] propose two open set SAR ATR methods based on generative model via random sampling combination (GvRSC). In detail, they mix up the training samples or deep features from different known classes to generate unknown class samples or deep features. However, given the complexity of the generation process and the data imbalance issue caused by excessive fake target generation, it is exceedingly challenging in practice to generate and learn enough fake targets to cover all possible anomalies that a classifier may encounter in real-world scenarios [32].

Most OSR methods based on discriminative model adopt the threshold-based recognition scheme, where the closed decision boundaries are usually determined via the preset reject threshold. Previously, there are initial attempts involving shallow discriminative models in the development of OSR methods, such as support vector machine (SVM) [16, 33, 34], sparse representation [35], nearest neighbor [36] and so on [37–40]. However, limited by their constrained feature extraction capabilities, shallow discriminative models perform poorly in terms of classification accuracy and reconstruction precision. Consequently, they are gradually being replaced by deep discriminative models [22]. Bendale et al. [41] adopt the deep learning technologies and introduce an OSR method where the SoftMax layer in the network is substituted with OpenMax, which uses Weibull distribution to recalibrate the output probabilities. Furthermore, Giusti et al. [24] evaluate the OpenMax on MSTAR dataset and the recognition results demonstrate that OpenMax reaches passable performance on some open set SAR ATR tasks. Oveis et al. [26] enhance OpenMax by incorporating different activation functions and class-specific parameters. The extended OpenMax proves to be more adept at open set SAR ATR. Oza et al. [42] propose C2AE, a method that introduces a novel reconstruction task aimed at enhancing the model's adaptability to the open set scenarios. Ma et al. [25, 43, 44] have conducted extensive research on SAR target recognition under open conditions. This

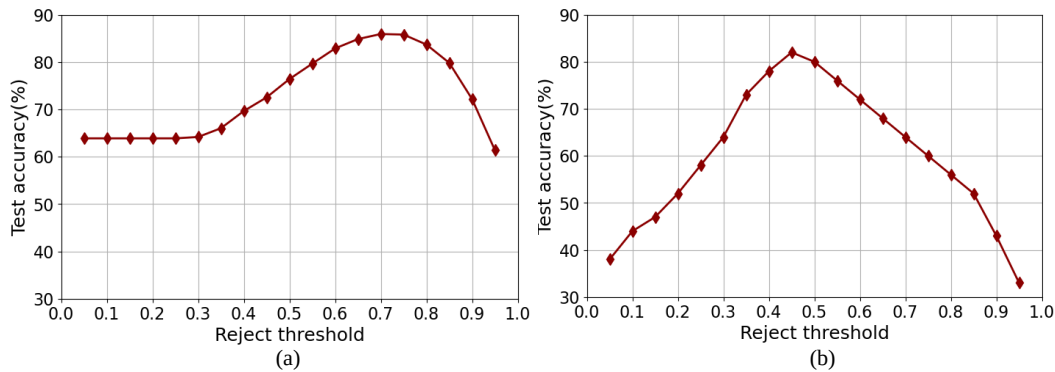


Fig. 2. The test accuracy at various reject thresholds on different datasets based on a plain CNN. (a) Results on the measured MSTAR datasets. (b) Results on the public CIFAR10 optical datasets. The classifier is trained on six target classes randomly selected from the training set respectively, and evaluated on the test set including six target classes included in the training set and four unknown classes at various reject thresholds. The test sample is rejected as unknown if the maximum output probability is below the preset reject threshold.

paper [25] proposes MultiGAN, which learns the known class distribution with GAN. Given that unknown class samples often deviate from the learned known class distribution, the model has the capability to reject unknown samples through the strategy of score thresholding, based on the discriminator's output. This paper [43] proposes a sub-dictionary learning process and employs extreme theory to determine the closed-set decision boundaries. To further update the classifier with these unknown targets, Ma et al. [44] combine the incremental learning strategy with the proposed open set classifier, which can not only reject targets from unknown classes but also continually learn to recognize unknown targets. Wang et al. [45] construct a dynamic feature space through the constructed meta-learning targets, which enhances the discrimination between known and unknown classes. To address the challenge of OSR in cross-domain scenarios, Zhao et al. [46] propose an open-set domain adaptation classification method, which rejects unknown classes by dynamically adjusting the threshold based on the weak alignment of the target domain. In recent years, prototype learning methods have demonstrated promising performance in OSR tasks. Notable examples of such methods include GCPL [47, 48] RPL [49], and ARPL [50]. These methods learn more compact intra-class features of training samples using prototypes, thereby reducing the potential risk of confusion between known and unknown classes. They can establish closed decision boundaries for known classes via a preset reject threshold on the distance space. It's important to highlight that these discriminative model-based OSR methods commonly encounter the task of finetuning the reject threshold in order to effectively discriminate between known and unknown classes.

It's a known fact that unknown class samples originate from an infinite space, encompassing multiple unknown classes. The impossibility of synthesizing samples for every unknown class stems from the lack of prior information within the training set. Conversely, discriminative model-based methods primarily concentrate on establishing closed decision boundaries for known classes, which is a task that is comparatively more attainable in practical applications. Therefore, this article focuses on the open set SAR ATR methods based on dis-

criminative model. Fig. 2 illustrates the accuracy of different datasets at various reject thresholds based on a plain CNN. As expected, the test accuracy is highly dependent on the reject threshold value, and the optimal reject threshold value for a specific OSR task may be not suitable for another OSR task. The results in Fig. 2 further demonstrate the importance of setting an accurate and adaptive reject threshold. Some studies have recognized the importance of the reject threshold and have attempted to set an accurate reject threshold in open set SAR ATR [51–53]. Inkawich et al. [51] preset a cut-off threshold to ensure a 95% true positive rate, which ensures high performance in recognizing known class samples. Liao et al. [52] determine the reject threshold based on the mean and standard deviation of the reconstruction errors of known class samples. Chen et al. [53] select the reject threshold by calculating the best F1-score on the validation set that only contains known class samples. Unfortunately, it is challenging to guarantee the capability of rejecting unknown targets without any prior knowledge of unknown classes.

To set a more accurate reject threshold, we should further consider the relationship between the distributions of known and unknown classes. Therefore, we propose an OSR method named ECTA with applications to SAR ATR. The encoder in the encoding module learns the latent representation of the training samples, which is then fed to the classification part to predict the class labels. The reconstruction strategy is designed in the conditional decoding module to improve the learning of network and approximate the reconstruction error distributions of known and unknown classes. Specifically, test samples from known classes will match one of the class labels included in the training set, whereas test samples from unknown classes will not match any of the class labels. Considering the above prior knowledge, the decoder is trained to accurately reconstruct matching samples and ensure that nonmatching samples are dissimilar to any known class sample. The reject threshold calculation module utilizes EVT to fit the maximum extreme value distribution of matching reconstruction errors and minimum extreme value distribution of nonmatching reconstruction errors. Finally, we obtain optimal reject thresholds adaptively by minimizing the probability

of false positives and false negatives. Here, false positives refer to the number of nonmatching reconstructed samples misclassified as known classes, and false negatives refer to the number of matching reconstructed samples misclassified as unknown classes. The combined operation of the conditional decoding and reject threshold calculation modules together realizes the reject threshold adaptation.

The main contributions of our work can be summarized as follows:

- 1) **We propose an open set SAR ATR method based on ECTA, which enables the network to adaptively obtain the reject thresholds.** This allows our method to reject unknown classes and recognize known classes simultaneously, leading to distinct advantages over closed set SAR ATR methods, as our method is better suited for real world scenarios where both known and unknown classes coexist.
- 2) **We propose a conditional decoding module, in which the reconstruction characteristics of known and unknown class samples are fully considered.** The proposed module can effectively learn and approximate the reconstruction error distributions of known and unknown class samples only using the training samples and class labels included in the training set.
- 3) **We design a reject threshold calculation module that aims to calculate the reject thresholds.** This module employs a precise distribution function fitting for the tails of reconstruction errors and calculates reject thresholds for each known class based on a probability minimization criterion.

The subsequent sections of this article are organized as follows: Section. II introduces the fundamental concepts of the autoencoder network and extreme value theory. Section. III elaborates on the proposed ECTA method. Section. IV presents comprehensive experiments conducted on the MSTAR dataset along with relevant analyses. The conclusion of the entire article is provided in Section. V.

II. PRELIMINARY

In this section, we will introduce the preliminary knowledge related to the proposed method to help readers better understand this article.

A. Autoencoder Network

The core idea underlying autoencoder (AE) [54] is to capture the efficient latent representation of the input sample, allowing for the reconstruction of these samples as closely as feasible to the original input while preserving the crucial information inherent in the input data. As shown in Fig. 4, AE network consists of two main parts: 1) the encoder; and 2) decoder. Usually, the process of encoding and decoding is achieved through the following transformations:

$$\mathbf{z} = s(\mathbf{W} \cdot \mathbf{x} + \mathbf{b}) \quad (1)$$

$$\tilde{\mathbf{x}} = s(\mathbf{V} \cdot \mathbf{z} + \mathbf{g}) \quad (2)$$

where $\mathbf{x}$ is the input sample, $\mathbf{z}$ is the latent representation extracted by the encoder, $\tilde{\mathbf{x}}$ is the output reconstructed sample, s is a nonlinearity function, and the parameters of the AE include the weight matrices $\mathbf{W}$ and $\mathbf{V}$, along with the bias vectors $\mathbf{b}$ and $\mathbf{g}$. Utilizing the gradient descent approach, the parameter optimization of the AE involves minimizing the objective function, as expressed below:

$$L(\theta) = \min_{\theta} \sum_{n=1}^N \|\mathbf{x}_n - \tilde{\mathbf{x}}_n\|_2^2 \quad (3)$$

where N is the number of the training samples, $\theta = \{\mathbf{W}, \mathbf{V}, \mathbf{b}, \mathbf{g}\}$ is the parameters in the AE.

AE, with its remarkable capability to extract nonlinear features from the inputs and accurately reconstruct the input sample, has gained significant popularity due to its simple model architecture and ease of implementation. As a result, it has found widespread application in numerous fields [28, 42, 55]. It's noteworthy that when the AE is trained only with known class samples in the training set, the reconstruction error for the samples from these target classes would be low. At test time, when faced with an unknown class sample, the model would not accurately reconstruct the input, thereby resulting in a high reconstruction error. Therefore, some researchers [42] only use samples included in the training set for training, AE indicates if a given sample belongs to target classes included in the training set based on the reconstruction error that needs to be less than a preset reject threshold. This heuristic represents how the decision boundary between known classes and unknown classes is established.

B. Extreme Value Theory

EVT is a branch of mathematical-statistical modeling that aims to analyze the distribution of data with extreme values. It has been widely used in many applications such as railway track inspection, finance, OSR [22, 35, 37, 38, 41, 42] and others. In this subsection, we will introduce the relevant concepts of EVT. In detail, we adhere to the formulation of the Pickands-Balkema-de Haan, also known as the Generalized Extreme Value theorem (GEV) [56]. Given n independent and identically distributed samples $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ drawn from an unknown distribution $P(x)$, EVT shows that given a sufficiently large location value u , the probability of an observation exceeding u by $\mathbf{x} > u$ conditioned on can be well approximated by a function related to Generalized Pareto Distribution (GPD) [57], i.e.,

$$P(\mathbf{x} > x + u | \mathbf{x} > u) = 1 - G(x) \quad (4)$$

and

$$G(x) = 1 - [1 + \zeta \frac{x}{\mu}]_+^{-\frac{1}{\zeta}}, x > 0 \quad (5)$$

where $G(x)$ is the cumulative distribution function (CDF) of the GPD with $\mu > 0$ and $\zeta \in \mathbb{R}$ is the cut-off at zero operation. At the same time, the probability distribution function (PDF) of the GPD can be given as:

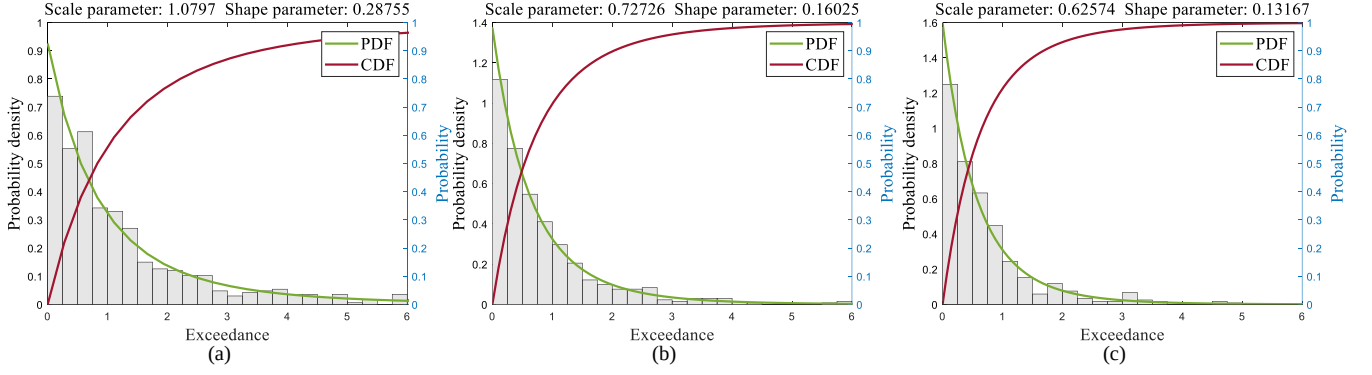


Fig. 3. Schematic of the fitted results of the pilot experiments, where the histograms represent the probability density of the exceedances, the green curves represent the fitted PDF, and the red curves represent the fitted CDF. In detail, we generate a large number of random values from the Student's t-distributions with (a) 3 (b) 5 and (c) 7 degrees of freedom respectively and then discard values less than 2 (a sufficiently large location value). Next, we subtract 2 from the unexcluded values to obtain the exceedances. Finally, we use the GPD to perform the probability function fitting to these exceedances and the estimated scale and shape parameters are shown on top of each figure.

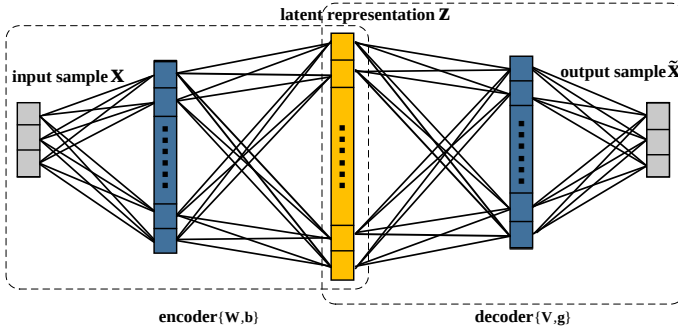


Fig. 4. Overview of the autoencoder network. The encoder projects the input sample to the latent space, and then the latent representation is fed to the decoder, which depicts the generative process from the hidden variable to data.

$$g(x) = \frac{1}{\mu} \left(1 + \zeta \frac{x}{\mu}\right)^{-\frac{1}{\zeta} - 1} \quad (6)$$

In statistics, the GPD is commonly employed to model the tails of sample distribution considering that it's difficult to select a probability distribution to model the entire sample distribution $P(x)$. GPD is usually specified by scale μ and shape ζ parameters, which can be conducted through the employment of the maximum likelihood estimation (MLE) technique based on the given observations. To validate the close fitness of the GPD to the extreme values, we perform several pilot experiments as shown in Fig. 3. It can be inferred from Fig. 3 that the fitted GPD is consistent with the extreme value distribution. As most of the discriminative information for the reject threshold is hidden in extreme value distributions [35], we further use the GPD to fit the extreme values of reconstruction errors. Through this operation, we can obtain a precise probability indicating the likelihood of an observation belonging to the fitted distribution based on the reconstruction errors.

III. PROPOSED METHOD

To better understand the OSR problems, we first give its common definition. Given the training samples $D_t =$

$\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)\}$ with K target classes, $y_i \in L_t = \{1, \dots, K\}$ is the class label of sample $\mathbf{x}_i$. The test set contains the target classes D_{known} included in the training set, and the unknown classes $D_{unknown}$ whose class label belongs to $K + 1$. OSR is to learn a function f only using D_t and L_t to achieve the goal:

$$\begin{cases} \mathbf{x}^j \in D_{known} \xrightarrow{f} j, & j \in 1, \dots, K \\ \mathbf{x} \in D_{unknown} \xrightarrow{f} K + 1. \end{cases} \quad (7)$$

where $\mathbf{x}^j$ denotes the test sample from the j th known class. The first row of Eq. (7) represents the accurate recognition for known classes, and the second row of Eq. (7) represents the precise rejection for unknown classes.

A. Network Architecture

The overall network architecture of ECTA, as shown in Fig. 5, consists of four parts: 1) the encoding; 2) classification; 3) conditional decoding; and 4) reject threshold calculation modules. The input of the encoding module is the input sample, and the output of the encoding module is the latent representation. Besides, the encoder in the encoding module contains the attention mechanism, which can improve the characterization ability of the extracted latent representation. The classification module incorporates a multi-layer perceptron (MLP) that takes latent representation as input and produces predicted labels as output. In the conditional decoding module, the novel random sampling reconstruction strategy is designed. In particular, the latent representation will be combined with the corresponding label and randomly combined with the label from other target classes to produce matching and nonmatching reconstructed samples through the decoder. The decoder learns to reconstruct matching reconstructed samples accurately while reconstructing nonmatching reconstructed samples dissimilar to any known class sample. The reject threshold calculation module includes two operations: MSE calculation and probability minimization criterion. After calculating the MSE for each training sample, this module partitions the set of matching and nonmatching reconstructed

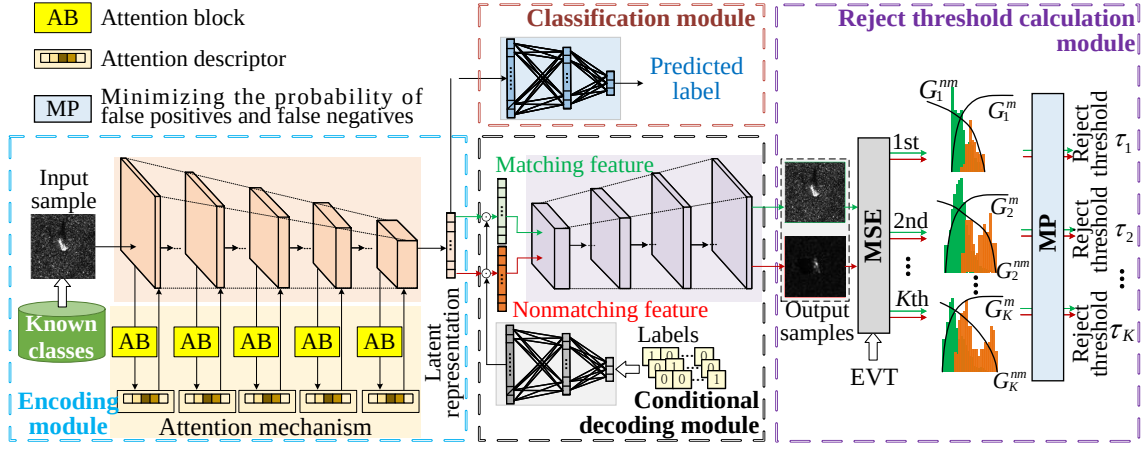


Fig. 5. Overview network architecture of ECTA. It consists of 1) the encoding module; 2) classification module; 3) conditional decoding module and 4) reject threshold calculation module.

reconstruction errors according to the class labels and then utilizes the GPD to perform the probability function fitting to the tails of both matching and nonmatching reconstruction errors. Optimal reject thresholds can be obtained via the probability minimization criterion. In summary, our method is capable of obtaining satisfactory reject thresholds for each known class, which avoids poor OSR performance and weak data adaptation that may arise from traditional OSR methods by manually presetting reject thresholds inaccurately.

1) Encoding module: The encoding module aims to produce representative latent representation for different target classes. The encoder refers to a plain CNN architecture [30] widely used in optical OSR methods with some changes in consideration of the characteristics of SAR images.

We replace the first few layers of 3×3 convolution kernels with 5×5 convolution kernels to obtain a larger receptive field (complete replacement across the network will significantly increase the number of parameters) as large receptive fields can mitigate speckle noise in SAR images to a certain extent [58]. At the same time, we apply the attention mechanism to the original input feature maps, improving the feature extraction ability in SAR images with speckle noise. The architecture of the attention block is given in Fig. 6. Given the input feature map $\mathbf{X}_{in} \in \mathbb{R}^{C \times H \times W}$, it will pass through the attention block to obtain the channel attention descriptor. In detail, an average pooling operation is first adopted along the spatial dimensions to produce a channel-level feature vector. The feature vector then passes through a convolution layer to reduce its channel dimension, followed by a rectified linear unit (Relu) activation function, another convolution layer to increase the channel dimension, and finally a sigmoid function to obtain the channel attention descriptor $\omega_c \in \mathbb{R}^{C \times 1 \times 1}$. The computation process is given as follows:

$$\omega_c = \text{Sigm}(\text{Conv}(\sigma(\text{Conv}(\text{AvgPool}(\mathbf{X}_{in})))))) \quad (8)$$

where Sigm represents the Sigmoid function, σ represents the Relu function. Then $\mathbf{X}_{in}$ is multiplied by ω_c to obtain the channel-wise refined feature map $\mathbf{X}_{out}$ as follows:

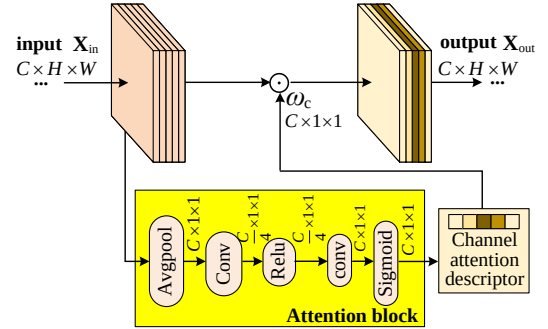


Fig. 6. Architecture of the attention block. The channel attention descriptor makes the network learn the importance of each feature channel in the original feature map.

$$\mathbf{X}_{out} = \omega_c \odot \mathbf{X}_{in} \quad (9)$$

where $\odot$ represents element-wise multiplication. The attention values of ω_c are broadcasted along the spatial dimensions. To ensure fairness, we apply the proposed encoding module to the compared methods in the Section. IV.

2) Classification module: The classification module aims to achieve accurate recognition for known classes. In particular, the classifier in the classification module is composed of an MLP. As mentioned above, the latent representation learned by the encoder is fed to the classifier and then the classifier produces the predicted label of the input sample. Given the set of training samples $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$ and their corresponding class labels $\{y_1, y_2, \dots, y_N\}$, the cross-entropy loss function is given as follows:

$$\mathcal{L}_1 = -\frac{1}{N} \sum_{i=1}^N \ell(C(F(\mathbf{x}_i)), y_i) \quad (10)$$

where C and F are the classifier and encoder respectively, ℓ is the cross-entropy loss function.

3) Conditional decoding module: As shown in Fig. 7, the input of the conditional decoding module is composed of two parts: the latent representation and class labels. The final

output of this module includes both matching and nonmatching reconstructed samples.

In this module, class labels are conditioned with the latent representation in line with the approach established by Perez et al. [59], which influences the latent representation $F(\mathbf{x}_i)$ by applying a feature-wise linear modulation $\mathbf{w}$ based on the conditioning information. To elaborate, the class labels are first transformed from the label space to the feature space using an MLP. Subsequently, the latent representation is combined with corresponding label vector and randomly combined with label vector from other target classes to produce matching and nonmatching reconstructed samples through the decoder, which can be given as follows:

$$\mathbf{w}_i^m = H(y_i^m), \quad \mathbf{w}_i^{nm} = H(y_i^{nm}) \quad (11)$$

$$\mathbf{z}_i^m = \mathbf{w}_i^m \odot F(\mathbf{x}_i), \quad \mathbf{z}_i^{nm} = \mathbf{w}_i^{nm} \odot F(\mathbf{x}_i) \quad (12)$$

$$\tilde{\mathbf{x}}_i^m = D(\mathbf{z}_i^m), \quad \tilde{\mathbf{x}}_i^{nm} = D(\mathbf{z}_i^{nm}) \quad (13)$$

where H represents several fully connected layers, D represents the decoder that mainly consists of several deconvolution layers, y_i^m represents the class label that matches the class identity of the input sample $\mathbf{x}_i$, y_i^{nm} represents the class label that nonmatches the class identity of the input sample $\mathbf{x}_i$, which is randomly selected from $L_t = \{1, 2, \dots, K\}$ and $y_i^{nm} \neq y_i^m$. $\tilde{\mathbf{x}}_i^m$ and $\tilde{\mathbf{x}}_i^{nm}$ represent the matching and nonmatching reconstructed samples respectively. Through the Eq. (12), we combine the label conditional information with the latent representation for subsequent learning by the conditional decoder. Considering the reconstruction characteristics of both known and unknown class samples, we design a random sampling reconstruction loss function as follows:

$$\ell^m = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i - \tilde{\mathbf{x}}_i^m\|_2 \quad (14)$$

$$\ell^{nm} = \frac{1}{N} \sum_{i=1}^N \|\mathbf{x}_i^{nm} - \tilde{\mathbf{x}}_i^{nm}\|_2 \quad (15)$$

$$\mathcal{L}_2 = \lambda \cdot \ell^m + (1 - \lambda) \cdot \ell^{nm} \quad (16)$$

where λ is a weight hyper-parameter. The loss function ℓ^m represents the reconstruction loss between matching reconstructed sample $\tilde{\mathbf{x}}_i^m$ and the input sample $\mathbf{x}_i$. The loss function ℓ^{nm} represents the reconstruction loss between nonmatching reconstructed sample $\tilde{\mathbf{x}}_i^{nm}$ and a random sampling sample $\mathbf{x}_i^{nm}$ from other target classes. During the inference, we feed the network with the most matching labels. The conditional decoder exhibits proficient reconstruction capabilities for known classes, yielding low reconstruction error. Conversely, if the test sample belongs to the unknown class and is dissimilar to any known target, the conditional decoder will not be able to reconstruct it well, resulting in a high reconstruction error. If the test sample belongs to the unknown class but closely resembles a known target, the conditional decoder will try to reconstruct it as $\mathbf{x}^{nm}$, inevitably leading to a high

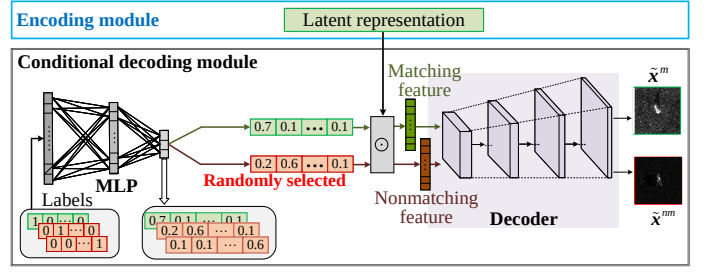


Fig. 7. Flow diagram of the conditional decoding module. The latent representation is combined with the corresponding class label to produce the matching reconstructed sample, and combined with the class label from a random other target class to produce the nonmatching reconstructed sample.

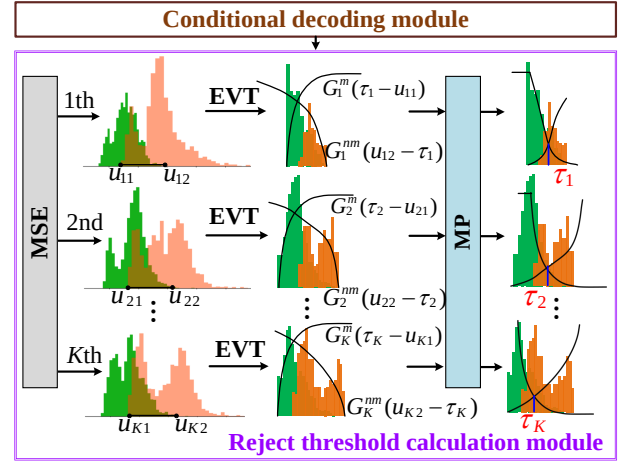


Fig. 8. Flow diagram of the reject threshold calculation module. It mainly consists of two operations: reconstruction error calculation (MSE) and probability minimization (MP) criterion.

reconstruction error.

4) Reject threshold calculation module: The reject threshold calculation module aims at obtaining optimal reject thresholds as shown in Fig. 8. Given the i th training sample, this module first calculates the MSE between the input sample and reconstructed sample, including the matching reconstruction error r_i^m and nonmatching reconstruction error r_i^{nm} . The set of matching and nonmatching reconstruction errors based on the training set can be given as follows:

$$CS^m = \{r_i^m | r_i^m = \|\mathbf{x}_i - \tilde{\mathbf{x}}_i^m\|_2\} \quad (17)$$

$$CS^{nm} = \{r_i^{nm} | r_i^{nm} = \|\mathbf{x}_i - \tilde{\mathbf{x}}_i^{nm}\|_2\} \quad (18)$$

where CS^m and CS^{nm} represent the set of matching and nonmatching reconstruction errors respectively. Then, this module further partitions the reconstruction errors r_i^m and r_i^{nm} into different subsets corresponding to the class label of the input sample $\mathbf{x}_i$. Finally, we can obtain $\{CS_1^m, CS_2^m, \dots, CS_K^m\}$ and $\{CS_1^{nm}, CS_2^{nm}, \dots, CS_K^{nm}\}$, where CS_j^m and CS_j^{nm} , $j = \{1, 2, \dots, K\}$ represent the j th class set of matching and nonmatching reconstruction error subsets respectively.

Based on the set of matching and nonmatching reconstruction errors for each known class, this module further calculates the optimal reject thresholds. It can be inferred from Fig. 9

that the optimal threshold typically lies within the overlap region between the known and unknown reconstruction error distributions. Such a reject threshold achieves good trade-offs between false positives and false negatives. As most of the discriminative information for the reject threshold is hidden in the tails of the known and unknown class reconstruction error distributions, and at the same time, the decoder can give the approximate description for them, therefore, fitting the tails of matching and nonmatching reconstruction errors is crucial in enabling the model to calculate the accurate reject thresholds. In order to provide a more accurate fitting of the overlap region where the reject threshold is situated, this module employs the GPD to fit the extreme value distributions of both matching and nonmatching reconstruction errors. It has been empirically demonstrated in Section. II that the GPD offers a close approximation to the data with extreme values. In detail, we take the minimum value in CS_j^{nm} for the j th known class as the location value u_{j1} and fit the right tail of CS_j^m as G_j^m . Similarly, we take the maximum value in CS_j^{nm} as the location value u_{j2} and fit the left tail of CS_j^{nm} as G_j^{nm} . It's noteworthy that the GPD is only defined for fitting the maximum values, therefore, we perform inverse transform as $CS_j^{nm} = -CS_j^{nm}$ before fitting G_j^{nm} of the left tail of CS_j^{nm} .

Referring to Eq. (4), the optimal reject threshold can be calculated by solving the following formula:

$$\begin{aligned} \tau_j^* &= \min[P_j(R^m > \tau_j | R^m > u_{j1}) \\ &\quad + P_j(R^{nm} < \tau_j | R^{nm} < u_{j2})] \\ &= \min[P_j(R^m > \tau_j - u_{j1} + u_{j1} | R^m > u_{j1}) \\ &\quad + P_j(R^{nm} < -(u_{j1} - \tau_j + u_{j2}) | R^{nm} < u_{j2})] \\ &= \min[P_j(R^m > \tau_j - u_{j1} + u_{j1} | R^m > u_{j1}) \\ &\quad + P_j(-R^{nm} > u_{j2} - \tau_j - u_{j2}) | -R^{nm} > -u_{j2})] \\ &= \min[1 - G_j^m(\tau_j - u_{j1}) + 1 - G_j^{nm}(u_{j2} - \tau_j)] \end{aligned} \quad (19)$$

where τ_j^* denotes the optimal reject threshold for the j th known class and its value range is $[u_{j1}, u_{j2}]$. As the threshold τ_j increases, the false negative rate $P_j(R^m > \tau_j | R^m > u_{j1})$ decreases while the false positives rate $P_j(R^{nm} < \tau_j | R^{nm} < u_{j2})$ concurrently increases. In order to minimize Eq. (19), we adopt a simple linear search algorithm to find the optimal reject threshold τ_j . By iteratively solving the Eq. (19) K times, this module effectively calculates the optimal reject thresholds for each known class, achieving good trade-offs between false negatives and false positives.

B. Model Training and Open Set Recognition

1) Model Training: The encoding, classification, conditional decoding modules follow an end-to-end approach, employing the stochastic gradient descent (SGD) methods to iteratively update the parameters. The overall loss is determined through a meticulous weighted summation of $\mathcal{L}_1$ and $\mathcal{L}_2$, denoted as

$$\mathcal{L} = \mathcal{L}_1 + \beta \cdot \mathcal{L}_2 \quad (20)$$

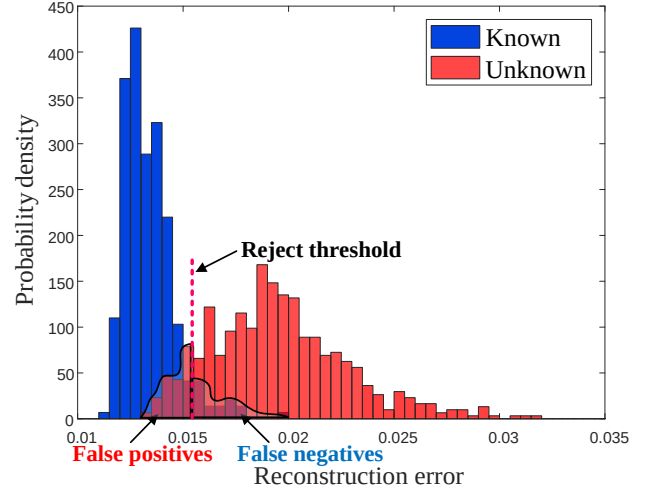


Fig. 9. The visualization of the probability density of known and unknown class reconstruction errors. If the reconstruction error is smaller than the reject threshold, the sample will be classified as a positive sample (Known), otherwise, it will be classified as a negative sample (Unknown). False positives denote the number of negative samples (Unknown) misclassified as positive samples (Known), and false negatives refer to the number of positive samples (Known) misclassified as negative samples (Unknown).

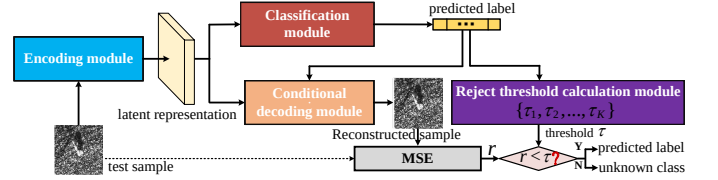


Fig. 10. Flow diagram of the test stage. The reconstruction error between the test sample and the reconstructed sample is compared to the class reject threshold to determine whether the test sample belongs to the unknown class or not.

where β is a weight hyper-parameter. The overall loss encapsulates the contributions of both components, effectively shaping and guiding the network's learning process.

Algorithm 1 shows the pseudo-code of the training stage of the proposed method.

Algorithm 1 The pseudo-code of the training stage

- 1: **Input:** the training data $D = \{(x_1, y_1), \dots, (x_N, y_N)\}$; the hyper-parameter β ;
- 2: **Output:** the trained network $F(\cdot)$, $C(\cdot)$, $H(\cdot)$ and $D(\cdot)$;
- 3: Obtaining the latent representation of the training sample $\mathbf{f}_i = F(\mathbf{x}_i)$;
- 4: Obtaining the predicted class label of the training sample $\mathbf{p}_i = C(\mathbf{f}_i)$;
- 5: Computing the cross-entropy loss $\mathcal{L}_1$ through Eq. (10);
- 6: Obtaining the reconstructed samples $\mathbf{x}_i^m$ and $\mathbf{x}_i^{nm}$ through Eq. (11)-(13);
- 7: Computing the conditional reconstruction loss $\mathcal{L}_2$ through Eq. (14)-(15);
- 8: Updating the network parameters until the training loss $\mathcal{L} = \mathcal{L}_1 + \beta \cdot \mathcal{L}_2$ converges

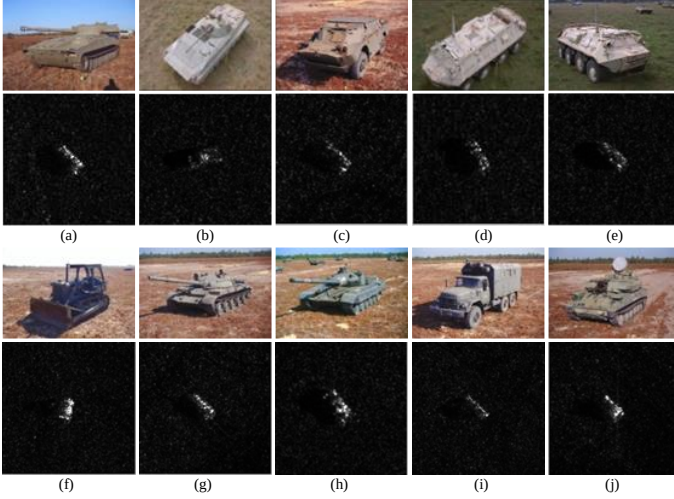


Fig. 11. Illustrations of ten distinct ground targets are provided, with optical images presented at the top and corresponding SAR images presented at the bottom. (a) 2S1. (b) BMP2. (c) BRDM2. (d) BTR70. (e) BTR60. (f) D7. (g) T62. (h) T72. (i) ZIL131. (j) ZSU234.

2) Open Set Recognition: Fig. 10 illustrates the test stage of ECTA, wherein the test sample undergoes a series of modules. The test sample is input to the encoding module to get the latent representation, which is fed to the classification module, yielding the predicted class label. Then, the predicted class label and the latent representation are input into the conditional decoding module to produce the reconstructed sample. Finally, a meticulous comparison is conducted between the reconstruction error of the reconstructed sample and the class reject threshold. If the reconstruction error is larger than the reject threshold, the test sample is rejected as the unknown class; otherwise, it's confidently recognized as the predicted label.

IV. EXPERIMENTS

A. Experimental Settings and Evaluation Metrics

1) Experimental Settings: The ECTA is verified with the MSTAR [60] public dataset, which is collected using the 10-GHz SAR sensor in 0.3×0.3 m resolution. MSTAR dataset contains ten types of vehicle targets and the examples of the ten different ground targets are shown in Fig. 11. Following the recommended setup, the network is trained using images with a depression angle of 17° , and tested on the images with a depression of 15° . The number of each class target image is shown in Table. I. Moreover, to further validate the effectiveness of ECTA, we conduct OSR experiments on the FUSAR-Ship dataset [61], which is collected from Gaofen-3 satellite. The resolution of the SAR images is 1.5m in range and azimuth. In this paper, we category several types of ship targets contained in the FUSAR-Ship dataset into four parts: cargo ships, container ships, fishing vessels, and oil tankers. Table. II shows the ship target numbers of each category in our paper. Fig. 12 shows the examples of the four different ground targets of ship targets.

Next, we will introduce the implementation details of our method. In Eq. (16) and (20), we set the balanced parameters

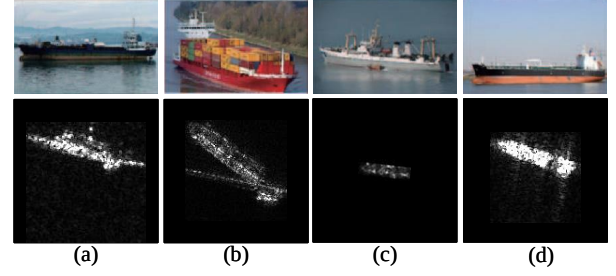


Fig. 12. Illustrations of four distinct ground targets are provided, with optical images presented at the top and corresponding SAR images presented at the bottom. (a) Cargo ships. (b) Container ships. (c) Fishing vessels. (d) Oil tankers.

λ and β as 0.5 and 1 respectively. We employ the stochastic gradient descent (SGD) optimizer for training, with a weight decay of 0.0005 and momentum of 0.9. A mini-batch size of 32 is used, along with an initial learning rate of 0.1. The network undergoes a total of 200 training epochs, and the learning rate is decreased by a factor of 10 at the 60th, 120th, and 180th epochs. These experiments are implemented with the PyTorch [62] deep learning framework.

2) Evaluation Metrics: We randomly select a part of target classes in the training set as known classes to train the network, and evaluate the OSR performance on the whole test set. The target classes not contained in the training stage are called as unknown classes. The openness [63] of the OSR tasks can be defined as

$$\mathbb{O} = 1 - \sqrt{\frac{2 \times N_{\text{train}}}{N_{\text{train}} + N_{\text{test}}}} \quad (21)$$

where N_{train} is the number of known classes included in the training set, N_{test} is the number of target classes included in the test set. The degrees of openness will change with the variation in the number of known classes N_{train} . In our experiments, we set $N_{\text{train}} = K$. Correspondingly, the number of the unknown classes is $10 - K$. To change the degree of openness, we vary K with a fixed $N_{\text{test}} = 10$.

To analyze the OSR performance quantitatively, we introduce recall, precision and F1 (F1-score) to evaluate different OSR methods. The formulas are as follows:

$$\text{recall}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FN}_i}, \quad \text{recall} = \frac{\sum_{i=1}^K \text{recall}_i}{K} \quad (22)$$

$$\text{precision}_i = \frac{\text{TP}_i}{\text{TP}_i + \text{FP}_i}, \quad \text{precision} = \frac{\sum_{i=1}^K \text{precision}_i}{K} \quad (23)$$

$$\text{F1}_i = \frac{2 \times \text{precision}_i \times \text{recall}_i}{\text{precision}_i + \text{recall}_i}, \quad \text{F1} = \frac{\sum_{i=1}^K \text{F1}_i}{K} \quad (24)$$

where TP_i (true positives) is the number of correctly classified targets of the j th known class, FP_i (false positives) is the number of targets misclassified as the j th known class, and FN_i (false negatives) is the number of targets of the j th known class misclassified as other target classes. The aforementioned approach ensures that each class contributes equally to the

TABLE I
DETAILED INFORMATION ON TEN TYPES OF VEHICLE TARGETS

Dataset	BMP2	BTR70	T72	BTR60	2S1	BRDM2	D7	T62	ZIL131	ZSU234
Training (17°)	233	233	232	256	299	298	299	299	299	299
Test (15°)	196	196	196	195	274	274	274	273	274	274

TABLE II
DETAILED INFORMATION ON FOUR TYPES OF SHIP TARGETS.

Dataset	Cargo ships	Container ships	Fishing vessels	Oil tankers
Training	233	233	232	256
Test	196	196	196	195

final F1, thereby providing a more comprehensive evaluation of performance across different classes [36, 64].

B. OSR Performance on the measured SAR Dataset

1) Result Comparisons under the same degree of openness on the MSTAR dataset: We select T72, BMP2 and BTR70 to treat as the training set. The test set includes three types of known class targets and seven types of unknown class targets. In detail, we compare ECTA with other OSR methods in Table. III, i.e., MultiGAN (SAR)[25], OsmIL (SAR)[38], C2AE [42], EVM [40], RPL [49], GCPL [47], GvRSC (SAR)[23], DIAS [30], OpenMax (SAR)[24] and SoftMax [55]. The baseline SoftMax stands for a plain network architecture where the network contains K classification output nodes, and unknown classes are rejected by thresholding the maximum output probability of the classifier. As for SoftMax, OpenMax, DIAS, GvRSC, GCPL and RPL, it's obvious that their OSR performance is all below 80%. OpenMax utilizes the Weibull distribution to calibrate the output probability. However, it lacks effective discriminative feature representation learning. As for GvRSC and DIAS, it's impossible to synthesize completely unknown class examples via the mixup strategy or a generative model, thus only reaching unsatisfactory OSR performance under such a large openness. GCPL and RPL improve the characterization ability of the extracted features using the prototype, but undergo the challenge of finetuning the reject threshold to distinguish between known and unknown classes.

OsmIL, C2AE, and EVM utilize the EVT to determine the closed decision boundaries for known classes, which reaches improved OSR performance compared to GCPL and RPL. However, in terms of the exemplar selection algorithm combined with EVT to find the closed decision boundaries for known classes, the closed decision boundaries determined by OsmIL or EVM are easily affected by the abnormal extreme points. As for C2AE, it trains the encoder and decoder separately, which may result in a situation where the encoder parameters are not well adapted to the decoder parameters. Therefore, the recognition performance of OsmIL, C2AE

TABLE III
OSR PERFORMANCE COMPARISON OF DIFFERENT METHODS WITH T72, BMP2 AND BTR70 BEING SELECTED AS KNOWN CLASSES. OPTIMAL PERFORMANCE IS DISPLAYED IN BOLD, AND SUB-OPTIMAL PERFORMANCE IS INDICATED WITH AN UNDERLINE.

Methods	OSR Performance (%)		
	Precision	Recall	F1-score
MultiGAN (SAR) [25]	<u>97.0</u>	98.2	97.5
OsmIL (SAR)[38]	87.0	93.4	89.9
C2AE [42]	89.1	86.9	87.3
EVM [40]	81.0	90.5	85.0
RPL [49]	78.8	74.7	78.2
GCPL [47]	72.9	83.7	77.0
GvRSC (SAR)[23]	67.8	73.2	69.2
DIAS [30]	61.5	79.3	69.1
OpenMax (SAR)[24]	67.8	76.0	67.5
SoftMax [55]	65.4	65.7	64.2
ECTA	98.2	<u>96.0</u>	<u>97.1</u>

and EVM is inferior to that of our method ECTA. From Table. III we can see that our method reaches superior OSR performance.

However, it's noteworthy that the OSR performance of ECTA is relatively lower compared to MultiGAN (F1-score: 97.1% vs. 97.5%). This discrepancy can be somewhat attributed to the fact that MultiGAN utilizes a linear search approach to finetune the reject threshold by considering the probability scores of all test samples, including those from unknown classes. It's important to highlight that this approach may not be entirely feasible in practice, as acquiring real unknown class samples to finetune the reject threshold can be challenging.

2) Result Comparisons under different degrees of openness on the MSTAR dataset: To further showcase the effectiveness of ECTA, we devise a comprehensive array of OSR experiments, encompassing various degrees of openness. Recognizing the significant impact of known and unknown class distribution differences on evaluation outcomes, we conduct five randomized trials with an identical degree of openness. In detail, we consider the following OSR methods to compare the OSR performance, including DIAS [30], C2AE [42], RPL [49], GCPL [47] and OpenMax [24]. To ensure the fairness, we apply the same modification as our method to the feature extractors of these methods. It can be inferred from Table. IV that with an increase in the degree of openness from 9.25% to 32.06%, the F1-score of all OSR methods experiences a decline. As for OpenMax, the OSR performance

TABLE IV
F1-SCORE AGAINST VARYING DEGREES OF OPENNESS WITH DIFFERENT METHODS. OPTIMAL PERFORMANCE IS DISPLAYED IN BOLD, AND SUB-OPTIMAL PERFORMANCE IS INDICATED WITH AN UNDERLINE.

Openess (%)	Methods	OSR trials 1-5 (%)					Average F1-score (%)
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5	
9.25	OpenMax	83.36	81.13	86.68	81.36	83.27	81.16
	GCPL	88.97	89.22	87.13	85.78	87.57	87.73
	RPL	86.61	76.76	83.90	83.16	85.15	83.12
	C2AE	<u>90.26</u>	84.90	86.29	85.13	87.22	86.76
	DIAS	89.72	<u>91.34</u>	91.31	<u>89.16</u>	92.69	90.84
	ECTA	91.98	91.73	<u>89.44</u>	90.55	<u>90.57</u>	90.85
13.40	OpenMax	74.01	67.94	81.92	72.76	73.71	74.07
	GCPL	82.82	78.62	83.82	82.33	84.32	82.38
	RPL	82.79	81.69	80.62	82.57	86.05	82.34
	C2AE	82.14	82.63	82.90	<u>84.12</u>	84.90	83.34
	DIAS	<u>83.74</u>	<u>84.18</u>	85.42	88.12	<u>89.90</u>	<u>86.27</u>
	ECTA	87.07	91.83	<u>84.80</u>	84.11	90.66	87.69
18.35	OpenMax	73.96	73.40	78.37	71.18	71.35	73.65
	GCPL	78.39	78.50	82.35	75.64	80.01	78.98
	RPL	73.37	75.35	86.31	<u>84.48</u>	86.00	81.10
	C2AE	82.83	72.75	84.03	70.20	82.31	78.42
	DIAS	<u>83.31</u>	<u>81.76</u>	83.13	79.22	<u>84.63</u>	<u>82.41</u>
	ECTA	85.62	83.07	<u>84.21</u>	86.85	84.50	84.85
24.41	OpenMax	72.32	73.52	71.07	59.57	63.69	68.03
	GCPL	67.01	<u>78.32</u>	68.05	67.66	69.37	70.08
	RPL	79.45	82.83	63.99	73.83	75.44	75.11
	C2AE	<u>85.35</u>	75.89	<u>85.67</u>	74.72	<u>78.27</u>	<u>77.98</u>
	DIAS	74.31	76.38	73.11	<u>76.82</u>	77.30	75.58
	ECTA	89.06	77.82	91.47	79.36	79.34	83.41
32.06	OpenMax	55.61	63.25	58.41	65.46	62.12	60.97
	GCPL	59.55	66.74	60.74	61.06	73.03	64.22
	RPL	73.23	68.23	60.97	58.59	75.31	67.27
	C2AE	<u>74.07</u>	<u>76.36</u>	<u>66.66</u>	74.85	<u>76.52</u>	<u>73.69</u>
	DIAS	70.26	73.97	59.54	<u>75.55</u>	74.25	70.71
	ECTA	77.08	81.33	75.27	78.58	78.09	78.07

to SAR images has almost been lost under such a large openness. However, the F1-score of ECTA degrades gently as the openness increases. These results suggest our method is capable of dealing with these open set scenarios with stable and exceptional performance.

Table. IV presents an intriguing observation that DIAS demonstrates outstanding OSR performance at a relatively low degree of openness: when the degree of openness equals 9.25%, DIAS reaches an impressive F1-score of 90.84%. However, when the degree of openness increases to 32.06%, the F1-score experiences a significant drop of approximately 20%, which is comparatively unsatisfactory when compared to the performance of our method. The main reason is that as the increase of openness, the number of unknown classes continues to increase, which significantly increases the difficulty of mimicking unknown class patterns for the generative model. As a consequence, the challenge of exhaustively spanning

the infinite open space becomes more pronounced for the generated examples as the increase of openness. For our method, the conditional decoding module produces a diverse set of nonmatching features by randomly combining the latent representation with the class labels. Even in situations with a limited number of known class samples, such a random combination approach still produces a significant number of nonmatching features to simulate the unknown patterns. These substantial nonmatching features play a vital role in describing the reconstruction characteristics of unknown classes. This interesting finding confirms our method is capable of dealing with a large number of unknown classes.

Fig. 13 (a)-(f) and Fig. 13 (a)-(f) show the visualizations of the confusion matrix and feature t-SNE of the test samples learned by “OpenMax”, “GCPL”, “RPL”, “C2AE”, “DIAS” and our method. Through the confusion matrix and feature t-SNE of the “OpenMax”, we can see that the known and

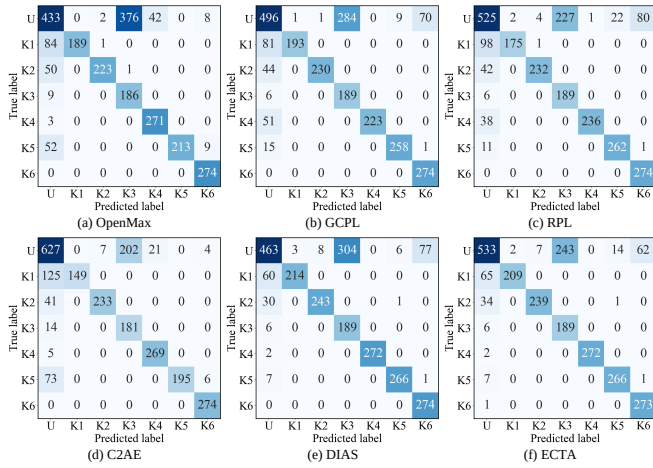


Fig. 13. The visualization of the confusion matrix for the “OpenMax”, “GCPL”, “RPL”, “C2AE”, “DIAS” and our method on the MSTAR dataset when the degree of openness equals 13.40%. The matrix is organized such that each row represents the true category while each column corresponds to the predicted category, where index “U” denotes the unknown class category and indexes “K1”-“K6” denote different known classes.

unknown class features have a large confusion, making the network cannot perform well on separating them. The utilization of prototype learning in “GCPL” and “RPL”, and generative model in “DIAS” significantly reduces the confusion for the known and unknown class features. Notably, our method attains higher accuracy rates for the test samples, indicating that the features learned by our approach are more distinguished for the different target classes. From Fig. 15 we can see that our method reaches better results for inter-class separation between known and unknown classes.

3) Result Comparisons under the same degree of openness on the FUSAR-Ship dataset: In this subsection, we further validate proposed method on the four types of ship targets. We randomly select three types of ship targets to treat as the training set. The test set includes three types of known class targets and one types of unknown class targets. In detail, we conduct three randomized trials to evaluate the average recognition performance. Table. V shows the OSR experiments on the four types of different methods. We choose OpenMax, GCPL, RL, C2AE and DIAS as our comparative OSR methods. The proposed method ECTA achieves a precision of 83.1% while ensuring high accuracy in recognizing known targets (with a recall exceeding 90%). This results in a final F1-score exceeding the comparative method by 2.1%. The aforementioned OSR experiments based on ship targets further validate the effectiveness of the proposed method.

C. Model Analysis

1) Analyzation of Conditional Decoding Module: The random sampling reconstruction strategy is proposed in the conditional decoding module to approximate the reconstruction error distributions of both known and unknown classes. In this section, we conduct an OSR trial with a degree of openness equal to 18.35% to verify the effectiveness of this strategy. We show the reconstruction results for training

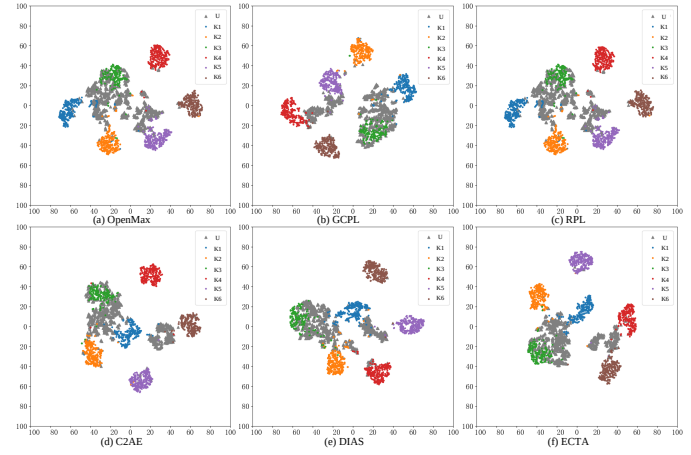


Fig. 14. The visualization of feature t-SNE of the test samples learned by the “OpenMax”, “GCPL”, “RPL”, “C2AE”, “DIAS” and our method. In detail, the input of the t-SNE is the latent representation obtained by the encoding module. Each digit in the legend denotes a target class, where index “U” denotes the unknown class category and indexes “K1”-“K6” denote different known classes.

TABLE V
OSR PERFORMANCE COMPARISON OF DIFFERENT METHODS WITH THREE RANDOMIZED TRIALS ON THE FOUR TYPES OF SHIP TARGETS.

Methods	OSR Performance (%)		
	Precision	Recall	F1-score
OpenMax (SAR)[24]	66.0	77.7	70.9
GCPL [47]	82.3	74.4	78.0
RPL [49]	87.6	73.9	79.7
C2AE [42]	83.1	82.8	82.4
DIAS [30]	<u>84.1</u>	<u>86.0</u>	<u>85.0</u>
ECTA	83.1	91.9	87.1

and test samples in Fig. 15 and the visualization results of the histogram of reconstruction errors and the calculated reject thresholds in Fig. 16. The observations from Fig. 15 reveal that the decoder demonstrates precise reconstruction capability for known class samples but lacks the capability to reconstruct unknown class samples, resulting in different reconstruction performance. Additionally, the decoder reaches similar reconstruction performance for both unknown class samples and nonmatching reconstructed samples, which indicates that the random sampling reconstruction strategy can give an approximate description for the reconstruction process of unknown class samples. From Fig. 16, we can see that the matching and nonmatching reconstruction error distributions give approximate descriptions of the known and unknown class reconstruction error distributions, which enables the network to calculate the optimal reject thresholds for each known class only using the training samples and class labels.

2) Analyzation of Reject Threshold Calculation: Based on the matching and nonmatching reconstruction errors for each known class, the reject threshold calculation module adopts probability distribution fitting and probability minimization criterion to calculate the optimal reject thresholds.

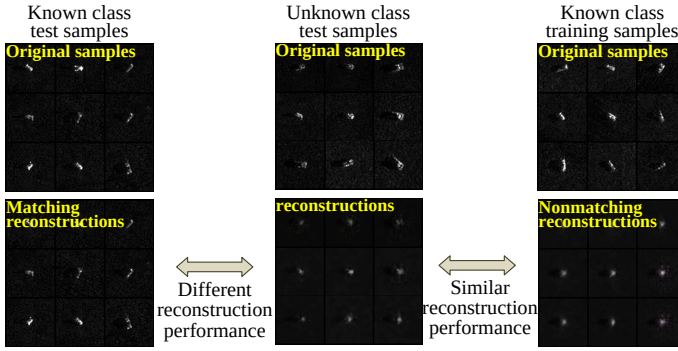


Fig. 15. The visualization of the reconstruction results for both the training and test samples. With the help of the random reconstruction strategy, the decoder approximates the unknown class reconstruction characteristic while effectively learning the known class reconstruction characteristic.

As shown in Fig. 16, due to the good approximation to the known and unknown class reconstruction error distributions and accurate probability function fitting, we can calculate the reject thresholds via minimizing Eq. (19). In summary, reject threshold calculation in ECTA allows for the acquisition of optimal reject thresholds for each known class, eliminating the need for manually tuning the reject threshold, thus addressing the inaccuracies and data adaptation challenges present in traditional OSR methods.

3) Ablation Study: In our proposed method, there are several innovative parts that affect the performance of open set SAR ATR, i.e., the encoding module, the conditional decoding, probability function fitting and target class split. To evaluate the impact of each part on OSR performance, we conduct several controlled OSR trials on the MSTAR dataset. In these trials, we keep the settings consistent except for the specific parts under examination. We conduct five randomized OSR trials and present the average F1-score in the Table. VI. We can see that all the proposed parts can improve the OSR performance. Row 1 in the Table. VI stands for an autoencoder with a branch of classifier, where the network rejects unknown class samples by thresholding the reconstruction errors. It should be noted that the encoder is a plain CNN architecture without proposed attention mechanism. When the proposed encoding module (row 2) is used for feature extraction, the F1-score further increases due to its superior feature extraction capability. The main reason why row 3 shows relatively good OSR performance is that compared to the original reconstruction loss in the autoencoder, the novel reconstruction loss in ECTA improves the learning of the network. Row 4 utilizes the EVT to fit the tails of matching and nonmatching reconstruction errors where the reject threshold is located, thus improving the accurate calculation of the reject threshold. Row 5 partitions the reconstruction errors according to the target class to achieve a more precise reject threshold calculation, thus the F1-score can be 2.88% higher than that without the target class split (row 3). Row 6 (our method) partitions the reconstruction errors and fits the probability function to calculate optimal reject thresholds for each known class, which reaches much better OSR performance. The results indicate that the four main parts of ECTA improve

TABLE VI
F1-SCORE BETWEEN DIFFERENT BASELINES FOR ABLATION ANALYSIS

Parts				F1-score (%)
Attention mechanism	Conditional decoding	Probability function fitting	Target class split	
-	-	-	-	73.50
✓	-	-	-	74.52
✓	✓	-	-	77.88
✓	✓	✓	-	80.93
✓	✓	-	✓	80.76
✓	✓	✓	✓	84.85

the OSR performance.

V. CONCLUSION

The main challenge of open set SAR ATR is that manually preset reject threshold is prone to inaccuracies, which is difficult to ensure the OSR performance and data adaptation. In response to this challenge, we propose a novel open set SAR ATR method named ECTA. ECTA captures the distinctive characteristics of both known and unknown class samples via the random sampling reconstruction strategy. Based on the matching and nonmatching reconstruction errors, ECTA utilizes EVT to fit the extreme value distributions and finally calculates the optimal reject thresholds for each known class by probability minimization criterion. The compelling results on the open set SAR ATR tasks demonstrate the effectiveness of the proposed method.

In the practical application of OSR methods for SAR images, unknown targets may belong to different classes. Simply rejecting unknown targets as the unknown class is insufficient. It is necessary to further distinguish their specific attributes. Therefore, in the future, we will focus on the discovery of novel class discovery (NCD) and class incremental learning (CIL). Building upon the rejection of unknown targets, NCD utilizes clustering techniques to discover different unknown classes within unknown targets. Next, we will further learn the discriminative information of new targets through the CIL strategy while retaining the ability to recognize old targets. By combining OSR, NCD, and CIL strategies, we can build a SAR ATR system with continuous learning capabilities, which is of great significance for the practical application of SAR ATR systems.

REFERENCES

- [1] A. Moreira, P. Prats-Iraola, M. Younis, G. Krieger, I. Hajnsek, and K. P. Papathanassiou, "A tutorial on synthetic aperture radar," *IEEE Geoscience and Remote Sensing Magazine*, vol. 1, no. 1, pp. 6–43, 2013.
- [2] O. Kechagias-Stamatis and N. Aouf, "Automatic target recognition on synthetic aperture radar imagery: A survey," *IEEE Aerospace and Electronic Systems Magazine*, vol. 36, no. 3, pp. 56–81, 2021.
- [3] Y. Sun, L. Lei, X. Li, X. Tan, and G. Kuang, "Patch similarity graph matrix-based unsupervised remote sensing change detection with homogeneous and heterogeneous sensors," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 6, pp. 4841–4861, 2021.

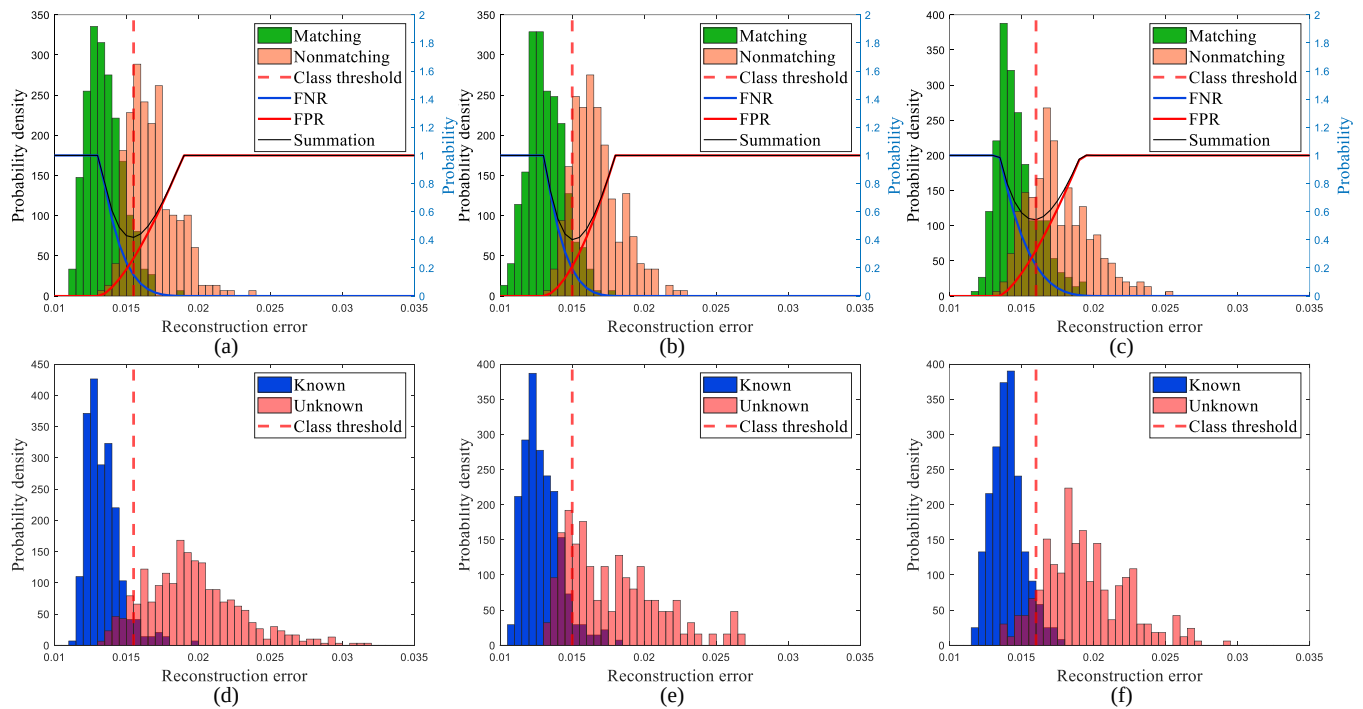


Fig. 16. The visualization of the histogram of reconstruction errors and the calculated reject thresholds in both the training stage (row 1) and test stage (row 2). In the training stage, we can obtain the false positive rate (FPR) and false negative rate (FNR) after probability function fitting, where FNR denotes the $P(R^m > \tau | R^m > u_1)$, FPR denotes the $P(R^{nm} < \tau | R^{nm} < u_2)$ and “Summation” denotes the probability of false positives and false negatives. Our method can calculate the optimal reject thresholds for each known class via a probability minimization criterion on the “Summation” curves.

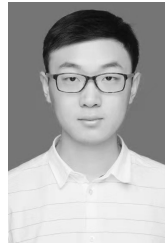
- [4] X. Hua, Y. Ono, L. Peng, Y. Cheng, and H. Wang, “Target detection within nonhomogeneous clutter via total bregman divergence-based matrix information geometry detectors,” *IEEE Transactions on Signal Processing*, vol. 69, pp. 4326–4340, 2021.
- [5] Y. Du, L. Du, and L. Li, “An sar target detector based on gradient harmonized mechanism and attention mechanism,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.
- [6] Y. Shi, L. Du, and Y. Guo, “Unsupervised domain adaptation for sar target detection,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 6372–6385, 2021.
- [7] Y. Shi, L. Du, Y. Guo, and Y. Du, “Unsupervised domain adaptation based on progressive transfer for ship detection: From optical to sar images,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–17, 2022.
- [8] S. Chen, H. Wang, F. Xu, and Y.-Q. Jin, “Target classification using the deep convolutional networks for sar images,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 54, no. 8, pp. 4806–4817, 2016.
- [9] X. Li, C. Li, P. Wang, Z. Men, and H. Xu, “Sar atr based on dividing cnn into cae and snn,” in *2015 IEEE 5th Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)*, 2015, pp. 676–679.
- [10] L. Du, L. Li, Y. Guo, Y. Wang, K. Ren, and J. Chen, “Two-stream deep fusion network based on vae and cnn for synthetic aperture radar target recognition,” *Remote Sensing*, vol. 13, no. 20, p. 4021, 2021.
- [11] Y. Li, L. Du, and D. Wei, “Multiscale cnn based on component analysis for sar atr,” *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–12, 2022.
- [12] C. Li, L. Du, Y. Li, and J. Song, “A novel sar target recognition method combining electromagnetic scattering information and gcnn,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.
- [13] D.-W. Zhou, H.-J. Ye, and D.-C. Zhan, “Learning placeholders for open-set recognition,” in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 4401–4410.
- [14] Z. Huang, X. Yao, Y. Liu, C. O. Dumitru, M. Datcu, and J. Han, “Physically explainable cnn for sar image classification,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 190, pp. 25–37, 2022.
- [15] Z. Huang, C. Wu, X. Yao, Z. Zhao, X. Huang, and J. Han, “Physics inspired hybrid attention for sar target recognition,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 207, pp. 164–174, 2024.
- [16] M. Scherrek and B. Rigling, “Multi-class open set recognition for sar imagery,” in *Automatic Target Recognition XXVI*, vol. 9844. SPIE, 2016, pp. 150–158.
- [17] M. D. Scherrek and B. D. Rigling, “Open set recognition for automatic target classification with rejection,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 52, no. 2, pp. 632–642, 2016.
- [18] J. Yang, K. Zhou, Y. Li, and Z. Liu, “Generalized out-of-distribution detection: A survey,” *arXiv preprint arXiv:2110.11334*, 2021.
- [19] M. Salehi, H. Mirzaei, D. Hendrycks, Y. Li, M. H. Rohban, and M. Sabokrou, “A unified survey on anomaly, novelty, open-set, and out-of-distribution detection: Solutions and future challenges,” *arXiv preprint arXiv:2110.14051*, 2021.
- [20] J. Liu, Z. Shen, Y. He, X. Zhang, R. Xu, H. Yu, and P. Cui, “Towards out-of-distribution generalization: A survey,” *arXiv preprint arXiv:2108.13624*, 2021.
- [21] J. Yang, P. Wang, D. Zou, Z. Zhou, K. Ding, W. Peng, H. Wang, G. Chen, B. Li, Y. Sun *et al.*, “Openood: Benchmarking generalized out-of-distribution detection,” *Advances in Neural Information Processing Systems*, vol. 35, pp. 32 598–32 611, 2022.
- [22] C. Geng, S.-j. Huang, and S. Chen, “Recent advances in open set recognition: A survey,” *IEEE transactions on pattern analysis and machine intelligence*, vol. 43, no. 10, pp. 3614–3631, 2020.
- [23] X. Geng, G. Dong, Z. Xia, and H. Liu, “Sar target recognition via random sampling combination in open-world environments,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 16, pp. 331–343, 2022.
- [24] E. Giusti, S. Ghio, A. H. Oveis, and M. Martorella, “Open set recognition in synthetic aperture radar using the openmax classifier,” in *2022 IEEE Radar Conference (RadarConf22)*. IEEE, 2022, pp. 1–6.
- [25] X. Ma, K. Ji, L. Zhang, S. Feng, B. Xiong, and G. Kuang, “An open set recognition method for sar targets based on multitask learning,” *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2021.

- [26] A. H. Oveis, E. Giusti, S. Ghio, and M. Martorella, "Extended openmax approach for the classification of radar images with a rejection option," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 59, no. 1, pp. 196–208, 2023.
- [27] Z. Ge, S. Demyanov, Z. Chen, and R. Garnavi, "Generative openmax for multi-class open set classification," *arXiv preprint arXiv:1707.07418*, 2017.
- [28] L. Neal, M. Olson, X. Fern, W.-K. Wong, and F. Li, "Open set learning with counterfactual images," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 613–628.
- [29] I. Jo, J. Kim, H. Kang, Y.-D. Kim, and S. Choi, "Open set recognition by regularising classifier with fake data generated by generative adversarial networks," in *2018 IEEE international conference on acoustics, speech and signal processing (ICASSP)*. IEEE, 2018, pp. 2686–2690.
- [30] W. Moon, J. Park, H. S. Seong, C.-H. Cho, and J.-P. Heo, "Difficulty-aware simulator for open set recognition," in *European Conference on Computer Vision*. Springer, 2022, pp. 365–381.
- [31] Y. Li, H. Ren, X. Yu, C. Zhang, L. Zou, and Y. Zhou, "Threshold-free open-set learning network for sar automatic target recognition," *IEEE Sensors Journal*, 2024.
- [32] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo *et al.*, "The many faces of robustness: A critical analysis of out-of-distribution generalization," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 8340–8349.
- [33] W. J. Scheirer, L. P. Jain, and T. E. Boult, "Probability models for open set recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 36, no. 11, pp. 2317–2324, 2014.
- [34] Q. Zhao and J. Principe, "Support vector machines for sar automatic target recognition," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 37, no. 2, pp. 643–654, 2001.
- [35] H. Zhang and V. M. Patel, "Sparse representation-based open set recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 39, no. 8, pp. 1690–1696, 2016.
- [36] P. R. Mendes Júnior, R. M. De Souza, R. d. O. Werneck, B. V. Stein, D. V. Pazinato, W. R. de Almeida, O. A. Penatti, R. d. S. Torres, and A. Rocha, "Nearest neighbors distance ratio open-set classifier," *Machine Learning*, vol. 106, no. 3, pp. 359–386, 2017.
- [37] S. Dang, Z. Cao, Z. Cui, Y. Pi, and N. Liu, "Open set incremental learning for automatic target recognition," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 57, no. 7, pp. 4445–4456, 2019.
- [38] S. Dang, Z. Cao, Z. Cui, and Y. Pi, "Open set sar target recognition using class boundary extracting," in *2019 6th Asia-Pacific Conference on Synthetic Aperture Radar (APSAR)*. IEEE, 2019, pp. 1–4.
- [39] B. Li, Z. Cui, Z. Cao, and J. Yang, "Incremental learning based on anchored class centers for sar automatic target recognition," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–13, 2022.
- [40] E. M. Rudd, L. P. Jain, W. J. Scheirer, and T. E. Boult, "The extreme value machine," *IEEE transactions on pattern analysis and machine intelligence*, vol. 40, no. 3, pp. 762–768, 2017.
- [41] A. Bendale and T. E. Boult, "Towards open set deep networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 1563–1572.
- [42] P. Oza and V. M. Patel, "C2ae: Class conditioned auto-encoder for open-set recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 2307–2316.
- [43] X. Ma, K. Ji, L. Zhang, S. Feng, B. Xiong, and G. Kuang, "Sar target open set recognition based on joint training of class-specific sub-dictionary learning," *IEEE Geoscience and Remote Sensing Letters*, 2023.
- [44] X. Ma, K. Ji, S. Feng, L. Zhang, B. Xiong, and G. Kuang, "Open set recognition with incremental learning for sar target classification," *IEEE Transactions on Geoscience and Remote Sensing*, 2023.
- [45] C. Wang, S. Luo, J. Pei, X. Liu, Y. Huang, Y. Zhang, and J. Yang, "An entropy-awareness meta-learning method for sar open-set atr," *IEEE Geoscience and Remote Sensing Letters*, 2023.
- [46] S. Zhao, Y. Zhang, Y. Luo, Y. Kang, and H. Wang, "Dynamically self-training open set domain adaptation classification method for heteroge-
- nous sar image," *IEEE Geoscience and Remote Sensing Letters*, 2024.
- [47] H.-M. Yang, X.-Y. Zhang, F. Yin, and C.-L. Liu, "Robust classification with convolutional prototype learning," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 3474–3482.
- [48] H.-M. Yang, X.-Y. Zhang, F. Yin, Q. Yang, and C.-L. Liu, "Convolutional prototype network for open set recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 5, pp. 2358–2370, 2020.
- [49] G. Chen, L. Qiao, Y. Shi, P. Peng, J. Li, T. Huang, S. Pu, and Y. Tian, "Learning open set network with discriminative reciprocal points," in *Computer Vision—ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part III 16*. Springer, 2020, pp. 507–522.
- [50] G. Chen, P. Peng, X. Wang, and Y. Tian, "Adversarial reciprocal points learning for open set recognition," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 11, pp. 8065–8081, 2021.
- [51] N. A. Inkawhich, E. K. Davis, M. J. Inkawhich, U. K. Majumder, and Y. Chen, "Training sar-atr models for reliable operation in open-world environments," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 3954–3966, 2021.
- [52] N. Liao, M. Datcu, Z. Zhang, W. Guo, J. Zhao, and W. Yu, "Analyzing the separability of sar classification dataset in open set conditions," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 7895–7910, 2021.
- [53] M. Chen, J.-Y. Xia, T. Liu, L. Liu, and Y. Liu, "Open set recognition and category discovery framework for sar target classification based on k-contrast loss and deep clustering," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024.
- [54] J. Masci, U. Meier, D. Cireşan, and J. Schmidhuber, "Stacked convolutional auto-encoders for hierarchical feature extraction," in *Artificial Neural Networks and Machine Learning—ICANN 2011: 21st International Conference on Artificial Neural Networks, Espoo, Finland, June 14–17, 2011, Proceedings, Part I 21*. Springer, 2011, pp. 52–59.
- [55] R. Yoshihashi, W. Shao, R. Kawakami, S. You, M. Iida, and T. Naemura, "Classification-reconstruction learning for open-set recognition," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 4016–4025.
- [56] J. Pickands III, "Statistical inference using extreme order statistics," *the Annals of Statistics*, pp. 119–131, 1975.
- [57] J. R. Hosking and J. R. Wallis, "Parameter and quantile estimation for the generalized pareto distribution," *Technometrics*, vol. 29, no. 3, pp. 339–349, 1987.
- [58] Z. Wang, L. Du, and Y. Li, "Boosting lightweight cnns through network pruning and knowledge distillation for sar target recognition," *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, vol. 14, pp. 8386–8397, 2021.
- [59] E. Perez, F. Strub, H. De Vries, V. Dumoulin, and A. Courville, "Film: Visual reasoning with a general conditioning layer," in *Proceedings of the AAAI conference on artificial intelligence*, vol. 32, no. 1, 2018.
- [60] DAPRA and AFRL, "The air force moving and stationary target recognition database." [Online]. Available: <https://www.sdms.af.mil/index.php?collection=mstar>
- [61] X. Hou, W. Ao, Q. Song, J. Lai, H. Wang, and F. Xu, "Fusar-ship: Building a high-resolution sar-ais matchup dataset of gaofen-3 for ship detection and recognition," *Science China Information Sciences*, vol. 63, pp. 1–19, 2020.
- [62] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "Pytorch: An imperative style, high-performance deep learning library," *Advances in neural information processing systems*, vol. 32, 2019.
- [63] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boult, "Toward open set recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 35, no. 7, pp. 1757–1772, 2012.
- [64] L. A. Jeni, J. F. Cohn, and F. De La Torre, "Facing imbalanced data—recommendations for the use of performance metrics," in *2013 Humaine association conference on affective computing and intelligent interaction*. IEEE, 2013, pp. 245–251.



Yiming Li received the B.S. and M.S. degrees in electronic and information engineering from Xidian University, Xi'an, China, in 2020 and 2023 respectively. He is currently pursuing the Ph.D. degree in signal processing at the National Key Laboratory of Radar Signal Processing of Xidian University.

His research interests include radar automatic target recognition and deep learning.



Yuchen Guo received the B.S. degree in electronic and information engineering and the Ph.D degree in signal processing from Xidian University, Shaanxi, Xi'an, China, in 2015 and 2021, respectively.

He is currently an Associate Professor with the Academy of Advanced Interdisciplinary Research, Xidian University. His research interests include radar automatic target recognition and pattern recognition.



Lan Du (Senior Member, IEEE) received the B.S., M.S. and Ph.D. degrees in electronic engineering from Xidian University, Xi'an, China, in Jul. 2001, Mar. 2004 and Jun. 2007 respectively.

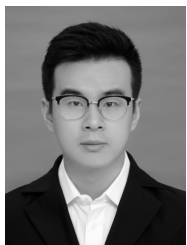
Her doctoral dissertation was granted National Excellent Doctoral Dissertation of PR China in 2009. From Sep. 2007 to Sep. 2009, she did research work as a Postdoctoral Research Associate in the Department of Electrical and Computer Engineering, Duke University, Durham, NC 27708 USA. She is currently a Professor at National Key Laboratory of

Radar Signal Processing, Xidian University. Her research work is supported by NSFC for Excellent Young Scholars. Her main research interests are in the fields of statistical signal processing and machine learning with application to radar target recognition.



Jian Chen received the B.Eng. degree in communication engineering from the Qingdao University of Technology, Qingdao, China, in 2014, and the Ph.D. degree in electronic engineering from Xidian University, Xi'an, China, in 2020.

He is currently a lecture at the National Key Laboratory of Radar Signal Processing of Xidian University. His research interests include radar signal processing, radar automatic target recognition, and pattern recognition.



Jialun Song received the B.Eng. degree in information counter technology from Xidian University, Xi'an, China, in 2019. He is currently pursuing the Ph.D. degree in signal processing at the National Key Laboratory of Radar Signal Processing of Xidian University.

His research interests include radar automatic target recognition and deep learning.



Zilin Wang received the B.Eng. degree in information counter technology from Xidian University, Xi'an, China, in 2020. He is currently pursuing the Ph.D. degree in signal processing at the National Key Laboratory of Radar Signal Processing of Xidian University.

His research interests include radar automatic target recognition and deep learning.